Shaffer began to study the long-term sustainability of this geographically isolated grizzly bear population.

Shaffer, a pioneer of population viability analysis, applied four factors that he considered would decide their fate. The first was demographic stochasticity: irregular, unpredictable fluctuations in numbers, age, gender, and birth and death rates. For example, if the overwhelming majority of animals in a population are males, breeding success will be poorer than in a more evenly balanced population, and will influence its chances of survival. The second consideration was environmental stochasticity: unpredictable fluctuations in environmental conditions, such as habitat and climate changes, which may affect the availability of food and shelter. The third was natural catastrophes, such as forest fires or floods. The fourth of Shaffer's factors was genetic changes, including problems created by inbreeding. For each of these, statistical modeling can determine a range of possibilities.

Since Shaffer's initial research in the 1970s and '80s, and subsequent new management and conservation strategies, grizzlies have extended their habitat by more than 50 percent within the extensive Greater Yellowstone Ecosystem—an area of 34,375 sq miles (89,031 sq km) that has the national park at its core. In 2014, the US Geological Survey estimated that around 757 bears lived in the ecosystem, based on 119 sightings of grizzly sows and cubs. That population reached an estimated 965 in 2022. Researchers monitoring the population, however, suggest that the population growth rate has been slowing since 2000. This suggests that the population may be reaching its maximum carrying capacity—the largest number of animals an area of suitable habitat can support. Nonetheless, grizzlies are still on the threatened species list.